

EXECUTIVE PERFORMANCE & DEMAND INTELLIGENCE DISCLOSURE

Executive Summary

This project presents an enterprise-grade Demand Intelligence System that transforms traditional inventory management into a data-driven decision framework. The solution integrates time-series forecasting, anomaly detection, and product clustering to optimize inventory planning, reduce stockouts, and improve warehouse efficiency. The system has been deployed as a live cloud-based dashboard for real-time decision support.

1. Time Series Forecasting & Demand Insights

Demand Pattern Analysis

Historical sales data (48 months) was analyzed using additive time-series decomposition, revealing three major components:

- Trend:** Monthly sales increased steadily from approximately \$34,000 (2015) to \$55,000 (2018), indicating sustained business growth.
- Seasonality:** Sales consistently decline during January–February and peak in November–December, reflecting strong holiday demand.
- Residuals:** Irregular spikes during late 2017 and 2018 suggest bulk institutional purchases and other exceptional events.

Forecast Model Comparison

Model	MAE	RMSE	MAPE
SARIMA (1,0,1)×(1,0,1) ₁₂	19,730.20	24,677.26	19.67%
Facebook Prophet	20,296.01	22,487.47	21.89%
XGBoost Regressor	29,364.30	29,366.84	32.78%

Production Model: SARIMA achieved the best overall forecasting accuracy and effectively captured seasonal demand, whereas XGBoost struggled to extrapolate long-term trends.

Key Forecast Insights

- Technology emerged as the fastest-growing category, with an **86.37%** demand increase from February to March.
- The **West** region showed the strongest growth, recording a **129.74%** increase during the same period.

Business Recommendation: Increase safety stock for Technology products in Western distribution centers before the expected March demand surge.

2. Operational Risk Monitoring & Inventory Optimization

Anomaly Detection

An Isolation Forest model identified **9 significant anomaly weeks**, primarily corresponding to:

- High-demand periods during **Black Friday** and **Cyber Monday**.
- Low-demand periods in **late December**.

A rolling **Z-Score** approach failed to detect these sustained seasonal changes, confirming Isolation Forest as the more reliable solution for operational monitoring.

Product Segmentation Using K-Means

Using the Elbow Method ($k = 4$), product sub-categories were grouped into four inventory strategies:

Cluster	Characteristics	Inventory Strategy	
0	High Value, Low Volatility (Copiers)	Just-in-Time procurement	
1	Low Volume, Stable Demand	Fixed Reorder Point (ROP)	
2	Growing Demand	Increase safety stock by 15–20%	
3	High Volume, High Volatility	Forecast-driven replenishment	dynamic

This segmentation enables optimized inventory allocation while reducing carrying costs and minimizing stockout risks.

System Deployment

The complete Demand Intelligence System has been successfully deployed as a cloud-hosted interactive dashboard for real-time forecasting and inventory monitoring.

Live Application:

<https://sales-forecasting-demand-intelligence-nvkpkku4xgcgbjeyhqdur6.streamlit.app/>